English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

The maximum total for all homework assignments during the semester is 50 points, corresponding to a bonus of 0.5 towards the exercise-class grade.

Problem 1. This problem examines in more detail the notion of a sigma-algebra on a set. Recall that a family $\mathcal{F}$ of subsets of a set Ω is a sigma-algebra (respectively, an algebra) on Ω if:

- $\Omega \in \mathcal{F}$;
- if $A \in \mathcal{F}$, then $A^c := \Omega \setminus A \in \mathcal{F}$;
- if $A_1, A_2, \ldots$ are countably many (respectively, finitely many for an algebra) elements of $\mathcal{F}$, then $\bigcup_i A_i \in \mathcal{F}$.

For brevity, "σ-algebra" below means "σ-algebra on Ω".

1. (2 points) Give an example of an algebra (on some Ω that you construct) that is not a sigma-algebra.
2. (2 points) Is every sigma-algebra closed under intersections of countably many of its elements?
3. (2 points) Prove that an intersection of σ-algebras is a σ-algebra. State whether you assume the intersection is finite, countable, or something else.

Let $\mathcal{G} \subset 2^\Omega$ be a class of subsets of Ω. Define the sigma-algebra generated by $\mathcal{G}$, denoted $\sigma(\mathcal{G})$, by

$$\sigma(\mathcal{G}) := \bigcap_{\substack{\mathcal{A} \text{ is a } \sigma\text{-algebra} \\ \mathcal{G} \subset \mathcal{A}}} \mathcal{A}.$$

4. (2 points) Prove that the definition is well defined, and that the above intersection is nonempty.

If Ω is a topological space, so that, roughly speaking, we can define $\mathcal{O}$ as the collection of open subsets of Ω, define the Borel σ-algebra on Ω by $\mathcal{B}(\Omega) := \sigma(\mathcal{O})$.

5. (2 points) Consider $\Omega = \mathbb{R}$. Is $\mathcal{O}$ countable or uncountable? Can we construct a countable $\widetilde{\mathcal{O}}$ such that $\mathcal{B}(\mathbb{R}) = \sigma(\widetilde{\mathcal{O}})$?
6. (2 points) Again let $\Omega = \mathbb{R}$. Prove that $\mathcal{B}(\mathbb{R})$ is generated by sets of each of the following forms, for $a, b \in \mathbb{R}$:
   (a) (a, b);
   (b) $[a, b]$;
   (c) $(a, b]$;
   (d) (a, ∞);
   (e) $[a, \infty)$.

Problem 2. (2 points) Which do you think is more likely: rolling at least n sixes with $6n$ dice, or at least $(n+1)$ sixes with $6(n+1)$ dice? Prove your answer.